

Load Interconnection Combined Feasibility / System Impact Study Report

For Queue Project Q11-0085

Snowden 138 kV
November 20, 2013

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q11-0085. The Interconnection Customer proposes a 173.2 MW large load (data center) facility interconnecting at the Snowden 138 kV substation in the Virginia area. The request was received on November 21, 2011. The study evaluated system intact and single contingency conditions using the regional planning case, including 60 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 11 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$42,457,000.

3. Network Impacts

3.1 Load Deliverability Analysis

Monitored Facility	Contingency	Loading (%)	Rating (MVA)	DFAX (%)
Reusens - Pentland 138 kV	Clover - Bear 345 kV L	136.0	208	72.2
Skimmer - Pentland 138 kV Line	Clover - Bear 345 kV	135.4	209	72.2
Snowden Substation - Skimmer 138 kV Line	Bear - Clover 345 kV	135.2	210	72.2
Balcony Falls - Butte 138 kV	Snowden - Skimmer 138 kV	125.4	162	100.0
Walerga-Galbrath - Aspen 138 kV line	Snowden - Skimmer 138 kV	110.9	204	100.0
Plaquemine Cogeneration Plant - Balconyfalls 138 kV line	Snowden - Skimmer Substation 138 kV	107.4	181	100.0
Walerga-Galbrath - Butte 138 kV	Skimmer - Snowden 138 kV L	106.3	201	100.0
Jack Ramey Wind - Starkey 138 kV	Reusens - Leesville 345 kV	102.6	250	5.9
Safeway Stores - Aseville 138 kV	Reusens - Leesville 345 kV	101.7	328	5.9
Boonsboro - Amago 138 kV line	Reusens - Leesville Substation 345 kV Line	101.5	241	14.5
Starkey - Hancock 138 kV	Leesville - Reusens Substation 345 kV	100.8	227	5.9

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Reusens - Pentland 138 kV	Rebuild line with higher-capacity conductor	\$12,001,000	89% (\$10,688,000)
Skimmer - Pentland 138 kV Line	Rebuild line with higher-capacity conductor	\$3,726,000	89% (\$3,318,000)
Skimmer - Snowden 138 kV	Rebuild line with higher-capacity conductor	\$4,419,000	89% (\$3,935,000)
Butte - Balcony Falls 138 kV	Rebuild line with higher-capacity conductor	\$12,031,000	100% (\$12,031,000)
Walerga-Galbrath - Aspen 138 kV line	Rebuild line with higher-capacity conductor	\$2,465,000	100% (\$2,465,000)
Plaquemine Cogeneration Plant - Balconyfalls 138 kV line	Reconductor line	\$1,841,000	100% (\$1,841,000)
Walerga-Galbrath - Butte 138 kV	Reconductor line	\$3,742,000	100% (\$3,742,000)
Jack Ramey Wind - Starkey 138 kV	Reconductor line	\$1,015,000	73% (\$739,000)
Safeway Stores - Asheville 138 kV	Reconductor line	\$4,151,000	73% (\$3,022,000)
Amago - Boonsboro 138 kV	Reconductor line	\$1,312,000	17% (\$220,000)
Starkey - Hancock 138 kV	Reconductor line	\$626,000	73% (\$456,000)
	Total allocated to project		\$42,457,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.